King Fahd University of Petroleum and Minerals
Computer Engineering Department

COE 510: Programing Methods for Robotics

Fall Semester 2026 (T261)

Project Proposal

Group Leader (Name/ID)

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Team (Student Names/ID):

2) SAUD ALMALKI g202016200

3) *N/A (two-member team)*

Format:

1. **Title:** A descriptive and concise **title** of your project indicating its **type**.

ROS 2-Based UGV and Multi-UAV System for Visual Monitoring and Autonomous Recovery

2. **Abstract:** A concise definition of the core of the project

This project aims to develop an integrated ground-and-aerial robotic system using ROS 2, Gazebo Harmonic, and RViz. A mobile unmanned ground vehicle carries up to eight unmanned aerial vehicles on a two-level carousel and transports them between designated survey locations. Selected UAVs deploy to search assigned areas, detect people and vehicles through simulated onboard cameras, and maintain target observations. An operator can select a detected target for subsequent visual following. UAVs return to the stationary carrier using AprilTag-guided alignment, with recovery confirmed through landing-pad contact, disarming, and latching.

An initial web-based prototype visualizes the concept and establishes the intended mission workflow. The main project will realize these capabilities through modular ROS 2 nodes, topics, services, actions, and coordinate transforms. Gazebo will simulate robot dynamics, sensors, contacts, and the deployment mechanism; PX4 software-in-the-loop will provide UAV flight control; and RViz will display robot states, trajectories, detections, and assignments. Evaluation will examine mission completion, visual tracking, recovery accuracy, fault handling, and scalability through reproducible scenarios.

3. **Overview:** General overview of the project (case study) including its relevance, importance.

The project investigates coordinated operation between a mobile ground carrier and multiple aerial robots. The carrier transports and deploys UAVs at survey locations, while the UAVs provide aerial observation and visual monitoring. This ground-and-aerial arrangement provides a practical case study for robotics programming because it combines navigation, perception, asynchronous communication, mission sequencing, and interaction between multiple robots.

Development follows two stages. Stage 1 uses a web-based prototype to explore the system concept, carrier layout, deployment sequence, target-selection workflow, and operator interaction. Stage 2, which forms the main COE-510 deliverable, implements the intended system as a complete ROS 2 application with Gazebo simulation and RViz visualization.

The ROS 2 architecture will separate responsibilities into modules for ground-vehicle control, carousel operation, UAV interfaces, perception, target management, fleet coordination, mission control, and visualization. Topics will carry sensor data and robot states, services will support discrete commands and operator decisions, and actions will represent longer-running tasks such as navigation and surveying. Coordinate transforms will maintain consistency between the world, carrier, aircraft, cameras, and landing pads.

The intended mission proceeds as follows:

1. The operator configures a route, deployment stops, aircraft allocations, and mission parameters.
2. The UGV follows the route and stops at a deployment location.
3. The carousel presents the required tray, and selected UAVs launch after readiness and release checks.
4. UAVs survey assigned regions and produce image-based observations of people and vehicles.
5. A target-management module associates observations and maintains a catalogue for operator review.
6. An operator-selected target can be followed by one UAV, with bounded reacquisition and return behavior when observations become uncertain or unavailable.
7. UAVs return to their assigned pads using AprilTag-guided visual alignment. Contact, landing status, disarming, and latch confirmation establish successful recovery.
8. The carrier proceeds after the deployed group has recovered.

Gazebo will represent the wheel-driven carrier, carousel joints, aircraft dynamics, cameras, landing contacts, and an environment containing buildings and moving targets. RViz will provide a unified view of robot models, coordinate frames, routes, aircraft trajectories, detections, target identities, assignments, and mission status.

The COE-510 contribution centers on implementing and integrating these behaviors through ROS 2. It includes modular software design, state machines, concurrent sensor processing, typed communication interfaces, coordinate transformations, multi-robot namespaces, parameterized launch configurations, and automated testing.

Validation will progress from individual components and one UAV to multiple UAVs and complete multi-stop missions. Evaluation scenarios will include absent targets, moving targets, occlusion, similar-looking targets, camera loss, and vehicle faults. Measurements will include mission completion, observation freshness, tracking continuity, following distance, docking success, alignment error, contacts, and computational performance.

The final submission will contain the ROS 2 workspace, Gazebo models and worlds, RViz configuration, launch files, documented interfaces, reproducible tests, validation results, and a demonstration.

4. **Plan:** A table listing the planned tasks to complete the project, tasks' deadlines, and responsible member(s).

The project schedule starts in **Week 9** and finishes in **Week 15**. Each deadline is the end of the stated course week.

Planned task	**Deadline**	**Responsible member(s)**
Document Stage 1 requirements; define the final ROS 2 architecture, interfaces, and measurable acceptance criteria.	Week 9	Both members
Establish ROS 2 packages, topics, services, actions, coordinate frames, parameters, and launch configurations.	Week 10	Mohammed
Integrate the Gazebo carrier, wheel control, carousel, latches, and contact sensors with ROS 2.	Week 11	Saud
Integrate PX4 and camera communication with ROS 2; verify single-UAV deployment, flight, and recovery; establish image detection and target observations.	Week 12	Both members
Integrate localization, target association, selected-target following, and AprilTag recovery sequencing; evaluate two-UAV coordination.	Week 13	Both members
Complete RViz visualization; scale toward eight UAVs and multi-stop missions; evaluate target loss, faults, docking, and processing performance.	Week 14	Both members
Complete system integration and regression checks; prepare the report, reproducible demonstration, and presentation.	Week 15	Both members

Member key

Mohammed: MOHAMMED ALI SHAKEEL ASHTIKAR (g202621080), group leader.

Saud: SAUD ALMALKI (g202016200).

Both members share system integration, validation, documentation, and the final demonstration. Gazebo model preparation and ROS 2 development can proceed in parallel.

5. **References/Resources:** A list of relevant references/resources (8 minimum) to be used.

[1] S. Macenski, T. Foote, B. Gerkey, C. Lalancette, and W. Woodall, "Robot Operating System 2: Design, Architecture, and Uses in the Wild," *Science Robotics*, vol. 7, no. 66, 2022. DOI: 10.1126/scirobotics.abm6074.

https://arxiv.org/abs/2211.07752

[2] Open Robotics, *ROS 2 Jazzy Documentation*. Resources for nodes, communication interfaces, launch configurations, and testing.

https://github.com/ros2/ros2_documentation/tree/jazzy

[3] Open Robotics, *Gazebo Harmonic Documentation*. Resources for robot models, worlds, physics, sensors, and ROS integration.

https://gazebosim.org/docs/harmonic/

[4] Open Robotics, *RViz for ROS 2*. Visualization of ROS data, robot states, and coordinate frames.

https://github.com/ros2/rviz/tree/jazzy

[5] Open Robotics, *tf2 Coordinate Transformation Library*. ROS 2 packages for maintaining coordinate transforms.

https://github.com/ros2/geometry2/tree/jazzy

[6] PX4 Development Team, *ROS 2 User Guide*. Resources for PX4 communication and integration with ROS 2.

https://docs.px4.io/main/en/ros2/user_guide

[7] Z. Ge, S. Liu, F. Wang, Z. Li, and J. Sun, "YOLOX: Exceeding YOLO Series in 2021," arXiv:2107.08430, 2021.

https://arxiv.org/abs/2107.08430

[8] Y. Zhang et al., "ByteTrack: Multi-Object Tracking by Associating Every Detection Box," 2022. Background for detection association and tracking.

https://arxiv.org/abs/2110.06864

[9] AprilRobotics, *AprilTag 3: Detection and Pose Estimation*. Visual fiducial detection for landing-pad identification and alignment.

https://github.com/AprilRobotics/apriltag